PHYSICS ADAPTLY

ENERGY – TRANSFERS & EFFICIENCY

KEY VOCABULARY

Energy store: e.g. kinetic, thermal, chemical, gravitational.

Dissipated: energy spread out to the surroundings (often as heat).

Efficiency: useful output energy ÷ total input energy.

Conservation: energy cannot be created or destroyed, only transferred.

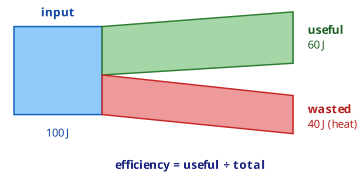
COMMON MISCONCEPTIONS

~~Energy gets used up.~~
✓ Energy is transferred and conserved, not used up.

~~Wasted energy disappears.~~
✓ Wasted energy is dissipated to the surroundings.

~~A device can be 100% efficient.~~
✓ Some energy is always wasted (usually as heat).

SANKEY: USEFUL VS WASTED



Arrow width shows the amount of energy; **efficiency = useful ÷ total**.

1 KINETIC ENERGY

A 2 kg ball moves at 4 m/s. **Calculate** its kinetic energy ($E_k = \frac{1}{2}mv^2$). **[2 marks]**

Answer: ______________________ J

2 EFFICIENCY

A motor takes in 100 J and usefully transfers 60 J. **Calculate** its efficiency. **[2 marks]**

Answer: ______________________ %

3 ENERGY STORES

A swinging pendulum transfers energy. Name the **two stores** involved. **[2 marks]**

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4 REDUCING WASTE

Explain how lubricating a machine reduces wasted energy. **[2 marks]**

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5 GPE

A 5 kg box is lifted 3 m (g = 9.8 N/kg). **Calculate** its gravitational PE. **[2 marks]**

Answer: ______________________ J

6 INSULATION

Complete the sentence using the word bank. **[2 marks]**

A wall with _____ insulation and a _____ thermal conductivity loses heat more slowly.

lower | thicker | higher | thinner

____

TIP: *Efficiency is always less than 1 (or 100%) — energy is never created.*

✓ Think: Where does the 'wasted' energy from a light bulb actually go?